MILITARY SPECIFICATION

FIELD CHANGE KIT FOR VARIABLE OUTPUT POWER CAPABILITY FOR TRANSMITTER AN/URT-23(C)

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the development and fabrication of a field change kit to the AN/URT-23(C) Transmitter (hereinafter referred to as the transmitter). The purpose of the field change kit is to permit the transmitter average power and peak envelope power (PEP) output level to be adjustable from 10 milliwatts (mW) to 1000 watts (W) (nominal).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbook. Unless otherwise specified, the following specifications, standards, and handbook of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901	Shock Test, H.I. (High-Impact); Shipboard Machinery, Equipment And Systems, Requirements For
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-W-16878	Wire, Electrical, Insulated, General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts; Packaging and Packing Of
MIL-F-17655	Field Changes And Field Change Kits; General Specification For
MIL-T-28706	Transmitter, Radio AN/URT-23(V)
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I - Environmental And Type II - Internally Excited)
MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-461	Electromagnetic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement Of
MIL-STD-965	Parts Control Program

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Electronic Systems Command (ELEX-8111), Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

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HANDBOOK

MILITARY

MIL-HDBK-251	Reliability/Design Thermal Applications

2.1.2 Other Government publication. The following other Government publication forms a part of this specification to the extent specified herein.

PUBLICATION

MILITARY

NAVAL SEA SYSTEMS COMMAND

Ship alteration (Shipalt)	SHIPALT Installation Instruction For SSBN 1152(K) PKG-F/A

(Copies of specifications, standards, handbook, and publication required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General. The field change kits shall be in accordance with MIL-E-16400 and Type I of MIL-F-17655, to the extent specified herein.

3.1.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3). The sample(s) shall be suitable for complete evaluation of mechanical and electrical form, design, performance, and ease of installation. The sample(s) shall be of final mechanical and electrical form, shall employ standard or nonstandard (when approved by the procuring activity) parts, and be completely representative of the final field change kit.

3.1.2 Parts, materials, and processes. Parts, materials, and processes shall be in accordance with MIL-E-16400, except as otherwise specified in 3.1.2.1 through 3.1.2.4. Particular attention shall be given to the acceptable and unacceptable materials of MIL-E-16400. Conformal coating of printed wiring boards is not required.

3.1.2.1 Parts control. The parts to be incorporated in the field change kit shall be controlled in accordance with MIL-STD-965, Procedure I.

3.1.2.2 Prior parts approval. Government-furnished equipment (GFE) may include nonstandard parts. Field change kits furnished in accordance with this specification shall employ standard parts or approved nonstandard parts. Nonstandard parts approvals granted under earlier contracts for equipments represented by the GFE shall not be construed as being extended to field change kits furnished in accordance with this specification. Nonstandard parts cited in Government-furnished drawings shall not be construed as being approved for use in field change kits furnished in accordance with this specification.

3.1.2.3 Individual selection. Field change kit performance shall not be dependent on the selection of individual parts, unless such selection is specifically approved by the procuring activity. If such individual part selection is approved, sufficient information shall be included in all applicable technical publications and engineering drawings to enable designated repair activities to make such selection.

3.1.2.4 Semiconductor devices. All semiconductor devices shall be made of silicon.

3.1.3 Interface requirements. The modification specified herein shall provide extended variable output capability as a supplement to the present normal modes of transmitter operation without degrading present normal mode performance. In addition, the installed hardware for this modification shall interface mechanically and electrically with the requirements specified in a and b:

a. Hardware installed by Shipalt SSBN 1152(K) PKG-F/A, which changes the nomenclature of the T-827( )/URT to T-1095/URT, and

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b. Hardware installed to adapt the transmitter to operate in the Naval Tactical Data System (LINK-11).

3.2 Government-furnished material (GFM). GFM, if any, shall be as specified (see 6.2).

3.3 Field change kits. Field change kits shall be designed and fabricated in accordance with MIL-F-17655, in the quantities specified in the contract (see 6.2).

3.4. Operating power requirements. All operating primary power requirements shall be derived from alternating current voltage within the transmitter. The installed units and piece parts may or may not derive their direct current power from existing power supplies within the transmitter.

3.4.1 Transient operation. The field change kit shall conform to the performance limits of operation under the conditions of transient voltage and transient frequency as specified in MIL-E-16400 for shipboard equipment, to the extent that performance of the GFE is not degraded.

3.5 Environmental service conditions. When the field change kit is installed, the transmitter shall be capable of continuous, reliable operation, within the performance limits specified in a through e, under those environmental conditions of MIL-E-16400 specified herein. The transmitter shall operate within tolerances throughout the corresponding tests specified in Section 4 without alignment or adjustment, other than external controls normally employed for operation of the equipment.

a. Operating temperature range (Range 4)

b. Nonoperating temperature range (Range 4)

c. Humidity (0 percent to 95 percent)

d. Shock

e. Vibration

3.6 Mechanical design and construction. Mechanical design and construction shall conform to all applicable requirements of MIL-E-16400, to the extent of being compatible with the GFE transmitter.

3.6.1 Accessibility. In addition to the requirements of MIL-E-16400, circuit mounting boards, components, test points, and maintenance adjustment components shall have maximum accessibility for maintenance, repair, or replacement. For modifications to existing assemblies, maintenance accessibility shall not be degraded, relative to the GFE. In new assemblies, all adjustable components shall be accessible without removing dust covers or other components.

3.6.2 Test provisions. A sufficient quantity of test points shall be provided and located to permit the isolation of malfunctioning parts or subassemblies within any assembly, without removing the printed circuit board from its connector or unsoldering any interconnection or wiring lead.

3.6.3 Factory adjustment controls. The design of the field change kits shall not include factory or sealed adjustment controls, unless specifically approved by the procuring activity. Where these controls are approved, a complete description and instructions for adjustment of the controls shall be included in the associated technical publications.

3.6.4 Shock and vibration.

3.6.4.1 Shock. The transmitter, with the field change kit installed, shall be capable of withstanding the shock test for Grade A, deck-mounted, Class II, lightweight and medium weight, as required, Type A equipment as specified in MIL-S-901.

3.6.4.2 Vibration. The transmitter, with the field change kit installed, shall be capable of withstanding the Type I vibration test of MIL-STD-167-1.

3.6.5 Printed wiring. The fabrication of printed wiring boards and the insertion of parts and assemblies thereon shall conform to the requirements of MIL-E-16400. All printed wiring interfacial (feed-through) connections on newly-designed double-sided boards shall be made by plated-through holes complying with the requirements of MIL-STD-275. Contractor certification requirements of MIL-P-55110 do not apply.

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3.6.6 Soldering. Soldering shall conform to the requirements of MIL-E-16400.

3.6.7 Attachment of wires and leads. Attachment of wires and leads shall conform to the requirements of MIL-E-16400.

3.6.8 Internal wiring and cabling. Internal wiring and cabling shall conform to the requirements of MIL-E-16400. All wiring shall be polytetrafluoroethylene insulated. Hook-up wire shall conform to MIL-W-16878, Type E.

3.6.9 Conductor identification. Identification of conductors shall conform to the requirements of MIL-E-16400. The method used for identification shall be consistent throughout the field change kit.

3.6.10 Part replacement. Part replacement shall conform to the requirements of MIL-E-16400.

3.6.11 Thermal design. Thermal design shall be in accordance with MIL-E-16400 to the extent of compatibility with the GFE. A thermal survey of the field change kit modifications shall be performed. MIL-HDBK-251 may be used as a guide.

3.6.12 Controls, indicators, and panel layout. The design of field change kit controls, indicators, and their arrangement on the front panel of the radio frequency (RF) amplifier shall be in accordance with MIL-E-16400.

3.7 Detailed performance. The modified transmitter shall conform to the requirements specified herein without degradation of any other parameters specified in MIL-T-28706, or without degrading the performance of the GFE.

3.7.1 General. The purpose of the field change kit is to extend the range of existing transmitter output power levels. The present design permits varying the output power from approximately 100 W to 1000 W. Incorporation of the field change kit developed in accordance with this specificaton will permit the transmitter output power to be varied continuously from 10 mW to 1000 W. The field change kit will also provide the capability to measure the forward and reflected output power over the extended range. All capabilities shall be provided within the existing form factor of the AM-3924( )/URT amplifier (see MIL-T-28706). Operation of the added power control circuitry shall preserve the amplitude relationships between continuous wave (CW), amplitude modulation (AM) carrier, and two-tone outputs in existing transmitters.

3.7.2 Power output.

3.7.2.1 Maximum output levels. The modified transmitter shall be capable of transmitting the power outputs specified in TABLE I into a 50-ohm resistive load on a continuously keyed basis throughout the 2 megahertz (MHz) to 30 MHz frequency range. The output level variation limits are based on setting the power amplifier front panel power level control for maximum output, and inserting standard two-tone audio inputs to the T-827( )/URT radio transmitter (see MIL-T-28706) in the single sideband (SSB), independent sideband, and compatible AM modes. In a transmitter equipped with the LINK-11 data modification, the audio input shall be the NET TEST signal at zero decibels referred to one milliwatt (dBm) average power and a PEP or average ratio of 8.6 decibels (dB) in each sideband.

3.7.2.2 Power output range. A power output level control located on the AM-3924( )/URT amplifier front panel shall permit the continuous adjustments of the output power to any desired level between 10 mW PEP, or average, and the full output levels of TABLE I, using multiple (coaxial) controls on a concentric shaft which replaces the existing control. The new control shall provide both power control and appropriate power monitor switching by suitable control and metering element ganging. For either normal or LINK-11 modified transmitters, only a single (coaxial) concentric control shall function to control power in all modes. All other power controls, and their panel legends, shall be removed.

3.7.2.3 Power output variation with temperature. Variation in power output shall not exceed a total of 1 dB over the 0° Celsius (C) to 50°C operating range for nominal power outputs of 100 W to 1000 W (maximum clockwise position of power control selector switch). For power settings of less than 100 W, variation shall not exceed ±1 dB (2 dB total) for the same temperature operating range.

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TABLE I. Power output.

Emission	Output level	Variation limits 1/
A1, F2j	1000 W average	0, +1 dB
A3h	250 W carrier with no modulation 1000 W PEP with modulation	0, +1 dB
A3a, A3j, A9b	1000 W PEP	0, +1 dB
A3b	1000 W PEP with two tones in each sideband	0, +1 dB
F9 (TADIL A 2/, double sideband) 3/	1000 W PEP, 100 W average	±1 dB
F9j (TADIL A, SSB) 3/	1000 W PEP, 200 W average	±1 dB

1/ Variation with output frequency only. Temperature and line voltage constant.

2/ Tactical digital information links.

3/ LINK-11 modified units.

3.7.2.4 Power output variation with supply voltage. For operation in the 100 W to 1000 W range, variation in power output shall not exceed 0.5 dB total over the range of steady-state line voltages and frequencies specified in MIL-T-28706. For operation below 100 W, variation in power output over the same steady-state band of supply voltages and frequencies shall not exceed 6 dB (total).

3.7.3 Power control distortion. RF output distortion (excluding distortion due to intermodulation and harmonics) generated by interaction of average power control, peak power control, and power amplifier closed loop power control circuits shall:

a. Produce distortion products reduced at least 30 dB below either tone of a two-tone output at 1000 W PEP

b. Produce not greater than 5 percent AM of a CW output at 1000 W

3.7.4 Keying and modulation transients. Transients shall conform to the requirements specified in a and b:

a. Keying transients. No overshoot shall be present in the initial keyed envelope in the data mode. In the normal (nondata) modes, overshoot on initial keying shall be such that the ratio of the maximum to the steady-state envelope voltages shall not exceed 1.25:1. The keyed envelope shall not exhibit ringing.

b. Modulation transients. No overshoot shall be present in the data mode on initial application of modulating signals. In the normal (nondata) modes, overshoot on initial application of modulation shall be such that the ratio of the maximum to the steady-state envelope voltages shall not exceed 3.0:1 for power control ranges below 100 W. No ringing shall occur on the modulated envelope. These requirements are applicable at any setting of the power control in any normal operating mode.

3.7.5 Power output monitoring. The output power monitor of the AM-3924( )/URT amplifier shall provide full scale forward or reflected power output indications of 150 mW to 1500 W in 10-dB increments (5 ranges). Reflected power monitoring shall also be possible at full scale indication of 0.1 of the full scale forward power metering range selected. Selection of FORWARD POWER, REFLECTED POWER, or REFLECTED POWER X 0.1 shall be by a toggle switch with spring return to the FORWARD POWER position. Meter indication shall be PEP.

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3.7.5.1 Metering accuracy. Forward power delivery to a 60-ohm load shall be within ±1 dB of the indicated value when measured at the 1000 W, 100 W, 10 W, 1.0 W, and 0.1 W points on the appropriate meter range providing upscale indication. Maximum total meter error shall not exceed 8 dB at a forward power indication of 10 mW into a 50-ohm load. Reverse power meter indications for a 50-ohm load shall not correspond to a voltage standing wave ratio greater than 1.4:1 at any operating frequency or power level.

3.7.5.2 Reverse power susceptibility. An available transmitter power of 100 mW at any frequency in the range from 40 MHz to 1000 MHz, applied from a 50-ohm source to the transmitter antenna terminals, shall not produce forward or reflected power indications greater than 5 mW.

3.7.6 Electromagnetic compatibility (EMC). The field change kit shall conform to the requirements of MIL-STD-461 for Class A4 communication-emission equipment.

3.7.7 Transmitter parameters. The requirements for intermodulation distortion, harmonic suppression, spurious emissions, noise, and hum distortion shall be as specified in MIL-T-28706.

3.7.8 Workmanship. Workmanship shall be in accordance with MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of, but is not limited to, a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, b) Government product inspection to measure quality of the product to be offered for acceptance, and c) Government inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been taken or until conformance of the product to prescribed criteria has been demonstrated.

4.2 General inspection. Inspection shall be as specified in MIL-E-16400, except as otherwise specified herein.

4.2.1 Quality assurance terms and definitions. Quality assurance terms used in this specification shall be as defined in MIL-STD-109.

4.2.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection

1. In-process inspection (see 4.4.1)

2. Production inspection (see 4.4.2)

c. Inspection of preparation for delivery (see 4.9)

4.3 First article inspection. Unless otherwise specified (see 6.2), one GFE transmitter shall have the field change kit installed and be subjected to first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE II.

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TABLE II. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Production inspection
Surface examination	3.1.2	4.2	X	X
Parts and materials	3.1.2	4.2	X	X
Workmanship	3.7.8	4.8	X	X
Mechanical design	3.6	4.2	X	
Soldering	3.6.6	4.2	X	X
Conductor identification	3.6.9	4.2	X	X
Output range, functional	3.7.1	4.7.8		X
Power output	3.7.2.1	4.7.1	X	
Power output range	3.7.2.2	4.7.1.1	X	
Power control distortion	3.7.3	4.7.1.2	X	
Keying transient	3.7.4	4.7.2	X	
Modulation transients	3.7.4	4.7.2	X	
Power output monitoring	3.7.5	4.7.1.3	X	
Metering accuracy	3.7.5.1	4..7.1.4	X	
Reverse power susceptibility	3.7.5.2	4.7.1.5	X	
Intermodulation distortion	3.7.7	4.7.3	X	
Harmonic suppression	3.7.7	4.7.3	X	
Spurious emissions	3.7.7	4.7.3	X	
Noise	3.7.7	4.7.3	X	
Hum distortion	3.7.7	4.7.3	X	
Transient voltage	3.4.1	4.7.6	X	
Transient frequency	3.4.1	4.7.6	X	
Low temperature	3.5	4.7.4	X	
High temperature	3.5	4.7.4	X	
Humidity	3.5	4.7.4	X	
Shock	3.6.4.1	4.7.5	X	
Vibration	3.6.4.2	4.7.5	X	
EMC	3.7.6	4.7.7	X	

4.4 Quality conformance inspection.

4.4.1 In-process inspection. Each field change kit shall be subjected to the inspection specified in 4.8.

4.4.2 Production inspection. Production inspection shall be made on every field change kit offered for delivery. Production inspection shall comprise such examinations and tests as are necessary to prove the workmanship and reveal omissions and errors of the production process. Production inspection shall include the examinations and tests specified in TABLE II.

4.4.3 Production field change kits. Production field change kits shall be identical to the kits that have been subjected to, and have passed, the first article inspection (see 3.1.1 and 4.3).

4.5 Test facilities, chambers, and apparatus. Test facilities, chambers, and apparatus shall be in accordance with MIL-E-16400.

4.6 Performance of tests. Performance of tests shall be in accordance with MIL-E-16400 and as specified herein.

4.7 Test methods.

4.7.1 Power output. Power output measurements shall be made at the end and mid-band frequencies of each RF amplifier frequency band for A1, A3h, and A3j modes of operation to verify compliance with 3.7.2.1.

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4.7.1.1 Power output range. Tests shall be performed to verify compliance with the requirements of 3.7.2.2. Power output shall be measured at the extremes of each power control range; continuous control, including power setting overlap when range switching, shall be verified. Measurements shall be performed at the center of each AM-3924( )/URT amplifier frequency band, and in the A1, A3h (carrier only), A3j (standard two-tone modulation), F9j, and F9 modes. Measurements in the F9j and F9 modes shall be with the NET TEST modulation signal.

4.7.1.2 Power control distortion. Measurements shall be made to verify compliance with 3.7.3. Measurements shall be performed at each AM-3924( )/URT amplifier band center frequency.

4.7.1.3 Power output monitoring. Measurements shall be made to verify compliance with 3.7.5.

4.7.1.4 Metering accuracy. Measurements shall be made to verify compliance with the requirements of 3.7.5.1 at operating frequencies of 2.1 MHz, 17.1 MHz, and 29.9 MHz. Accuracy shall be measured in A1 and A3j (standard two-tone modulation) modes.

4.7.1.5 Reverse power susceptibility. Measurements shall be made to verify compliance with the requirements of 3.7.5.2 for AM-3924( )/URT amplifier 2 MHz to 2.5 MHz, 15 MHz to 18 MHz, and 28 MHz to 30 MHz bands. The transmitter shall be keyed in the A3j mode, but unmodulated, and the power controls adjusted for minimum output.

4.7.2 Transient suppression. Measurements shall be made to verify compliance with 3.7.4. Keying and modulation transients shall be measured at the transmitter high gain frequency in the CW A3j upper sideband (USB), and F9j (USB) modes at maximum and minimum power control settings for the 1000 W to 100 W and 1 W to 100 mW power ranges. Separate oscilloscope photographs shall be obtained of keying and modulation characteristics, with oscilloscope sweep triggered by keyline closure or application of modulation, as appropriate. Keying characteristics measured shall include those for initial keying and maximum required keying rates. Modulation characteristics shall be recorded for initial application of modulation, following a period adequate to permit gain control voltages to decay to minimum output levels.

4.7.3 Transmitter parameters. Measurements of the following transmitter parameters shall be made to ensure compliance with the applicable requirements of MIL-T-28706. Measurements shall be in accordance with referenced test paragraphs of MIL-T-28706, modified as specified in a through e:

a. Intermodulation distortion (see 3.7.7). Band edges, 2 MHz to 2.5 MHz, 16 MHz to 18 MHz, and 28 MHz to 30 MHz bands. Measurements at 1 kilowatt (kW) PEP, 1 W PEP, 10 mW PEP, for USB only.

b. Harmonics (see 3.7.7). Operating frequencies of 2.0000 MHz, 4.0000 MHz, 8.0000 MHz, and 29.9999 MHz. CW operation at 1 kW, 1 W, and 10 mW.

c. Spurious emissions (see 3.7.7). Operating frequencies of 4.5000 MHz, 9.0000 MHz, and 25.0000 MHz. Unmodulated AM carrier operation at 250 W, 0.25 W, and 25 mW.

d. Noise (see 3.7.7). Operating frequencies of 2.2500 MHz, 17.0000 MHz, and 29.000 MHz. Reference A3j (USB) outputs of 250 W, 0.25 W, and 25 mW.

e. Hum (see 3.7.7). Reference single-tone SSB outputs of 250 W, 0.25 W, and 25 mW.

4.7.4 Temperature and humidity. Tests shall be conducted in accordance with MIL-E-16400. The following measurements shall be made before and after the temperature and humidity tests, and during temperature operating tests at the end of the period of operation at 0°C and at the end of the period of operation at +50°C: power output in the A1 and A3j modes and intermodulation distortion at transmitter high and low gain frequencies. Measurements shall be performed at maximum and minimum settings of the analog power control in the 100 W to 1000 W, 10 W to 100 W, and 10 mW to 100 mW control ranges. During the temperature operating test, except during the measurement period, the transmitter shall be keyed continuously to produce rated CW output during operation at +50°C. During the last hour of the high temperature operating test, air flow produced by forced convection in the test chamber shall be eliminated. During this period, air temperature in the immediate vicinity of the equipment under test shall be controlled to +50°C by radiation-baffled heating and cooling sources, as required. This test is intended to simulate the worst-case shipboard thermal environment. During humidity tests, measurements shall be made at the end of the second and fifth cycles.

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4.7.5 Shock and vibration. Prior to, and after completion of shock and vibration tests, the test specified in a shall be performed:

a. Power output range measurements in accordance with 4.7.1.1. Measurements shall be performed in the A1 and A3j (USB only) modes for the 100 W to 1000 W and 10 mW to 100 mW power control ranges in the 2 MHz, 17 MHz, and 29 MHz bands.

During the vibration test, the transmitter shall be operated and monitored in accordance with the requirements of MIL-T-28706. During the exploratory vibration portion of the test, the transmitter power controls shall be set to produce maximum output in the 1 W to 10 W range. During the variable frequency and endurance portions of the test, operation shall be split equally between maximum output settings in the 100 W to 1000 W and 1 W to 10 W ranges.

4.7.6 Power output variation with supply voltage. Measurements shall be performed to verify compliance with the requirements of 3.7.2.4. The transmitter shall be operated at the low gain frequency and measurements shall be performed over the range of steady-state line voltages and frequencies specified in MIL-T-28706. Measurements shall be performed at maximum and minimum settings of the analog power control in the 100 W to 1000 W and 1 W to 10 W power control ranges. Operation shall be in the CW and USB (standard two-tone modulation) modes.

4.7.7 EMC test. Compliance with 3.7.6 shall be verified by tests performed in accordance with MIL-STD-462.

4.7.8 Output range, functional test. Each field change kit shall be connected to a simulated power source to ensure the functional output of the equipment.

4.8 General workmanship. The field change kit, including subassemblies and assemblies, shall be examined for workmanship and soldering during the fabrication and assembly process for conformance to the requirements of 3.1.2.

4.9 Inspection of preparation for delivery. Inspection shall be conducted to ensure conformance with the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified herein, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, packing, and marking specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The transmitter covered by this specification is intended for general purpose voice, CW, radio teletypewriter and data communications, and may be used for surface ship, submarine, and fixed shore installations.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Number of first article samples to be submitted if other than specified in 4.3

c. GFE and GFM (see 3.1.2.2, 3.2, 3.4.1, 3.6, 3.6.1, 3.6.11, 3.7, and 4.3)

d. Levels of preservation, packaging, packing, and marking (see 5.1)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item. The first article should consist of at least one unit. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, and approval of the first article.

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6.4 GFE and GFM. The contracting officer should arrange to furnish the GFE and GFM specified in 6.2.

6.5 Nomenclature. Nomenclature referred to herein and those assigned to the GFE units may be subject to change in any contract resulting in equipment produced in accordance with this specification.

Preparing activity:
Navy - EC
(Project 5820-N768(EC))

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